Regular polyhedra with equal adjacent parameters

Human Compression Project

Abstract

We recover the exact matrix model and universal presentation of Li–Zhang for a self-dual regular polyhedron of type $\{n, n\}$ and order $n2^n$, for every $n \geq 3$. The group is the augmentation module of an n-cycle extended by its dihedral symmetry group. The common theorem recognizing the translation subgroup supplies an explicit inverse from this model to the presentation. We also identify the distinguished marking with the earlier polygon instance of Montero–Ivić Weiss's halving construction, and state the parameter specializations, the exceptional value $n = 2$, and the necessary correction to a displayed multiplication convention.

1 Statement and provenance

Write D_{2n} for the dihedral group of order $2n$ and $E_n = \{v \in \mathbb{F}_2^{\mathbb{Z}/n\mathbb{Z}} : \sum_i v_i = 0\}$. We use the common paper [U], specifically its graph generation, complement twist and cyclic recognition results. Their hypotheses are verified below. We retain the standard correspondence between string C-groups and regular abstract polytopes [MS02, Section 2E].

Theorem 1. *For each integer $n \geq 3$, the presentation*

$$\begin{aligned} P_n = \langle r_0, r_1, r_2 \mid\ & r_0^2 = r_1^2 = r_2^2 = (r_0 r_2)^2 = 1, \\ & (r_0 r_1)^n = (r_1 r_2)^n = 1, \\ & ((r_1 r_2)^k (r_0 r_1)^k)^2 = 1 \quad (1 \leq k < n)\rangle \end{aligned}$$

is, with its distinguished generators, the group $E_n \rtimes D_{2n}$ described below. It is a string C-group of type $\{n, n\}$, has order $n2^n$, and has an involutory automorphism reversing the marking. Its regular abstract polyhedron is therefore self-dual and has exactly $n2^n$ flags.

This is the principal construction of [LZ25, Theorems 1.3 and 3.4]. The same polyhedron occurs in [MW20, Section 4, physical page 9]. We recover both its marking and the exact later presentation; we do not claim a new existence result. The higher-rank Conjecture 3.7 in [LZ25] is a separate question and is not used in this proof.

2 The marking and its geometry

Let $H = \langle a, b \mid a^n = b^2 = 1,\ bab = a^{-1}\rangle$ act on $\Omega = \mathbb{Z}/n\mathbb{Z}$ by $a(i) = i + 1$, $b(i) = -i$. Put $s = ab$ and $q = e_0 + e_1$. In the left semidirect product

$$G_n = E_n \rtimes H, \qquad (v, g)(w, h) = (v + gw, gh),$$

take

$$r_0 = (0, s), \qquad r_1 = (0, b), \qquad r_2 = (q, s). \tag{1}$$

Since $sq = q$, these are nonidentity involutions. Also $r_0 r_2 = (q, 1)$ and $r_0 r_1 = (0, a)$. They generate H and one edge translation of the cycle, so the common graph-generation corollary gives the whole group G_n.

Apply the common complement-twist lemma with base point 0:

$$\delta(v, h) = (v + e_0 + he_0, h).$$

It fixes r_1 and interchanges r_0, r_2. Thus both adjacent products have order exactly n. The two dihedral subgroups

$$A = \langle r_0, r_1 \rangle = \{(0, h) : h \in H\}, \qquad B = \langle r_1, r_2 \rangle = \delta(A)$$

intersect in the stabilizer of 0, namely $\langle r_1 \rangle$. The remaining maximal marked subgroup is $C = \langle r_0, r_2 \rangle = \{1, r_0, r_2, (q, 1)\}$. Inspecting its translation coordinates gives $A \cap C = \langle r_0 \rangle$; applying δ gives $B \cap C = \langle r_2 \rangle$.

Each of A, B, C is a rank-two C-group. If two proper marked subgroups lie in the same one, its intersection property applies. Otherwise choose two different maximal marked subgroups containing them. Their intersection is the subgroup generated by their common marker, as just proved. Intersecting with that subgroup inside each maximal group gives exactly the subgroup on the common index set. Intersections with the full group are tautological. This proves the full intersection property for the three generators. Their distant pair commutes, so the marking is a string C-group. The retained correspondence now gives the polyhedron and its full flag-regular group. The reversal δ supplies its duality.

The basis $e_0 + e_i$, $i \neq 0$, gives $|E_n| = 2^{n-1}$, hence $|G_n| = n2^n$. This count describes the group already identified; it will not be used to prove the universal presentation.

3 An inverse map for the presentation

In P_n put $x = r_0 r_1$, $y = r_1 r_2$ and $b = r_1$. The defining relations give

$$x^n = y^n = b^2 = 1, \quad bxb = x^{-1}, \quad byb = y^{-1}, \quad (y^k x^k)^2 = 1 \quad (1 \leq k < n).$$

In the model (1), the first two images are $(0, a)$, $(0, b)$, while

$$y = (e_0 + e_{-1}, a^{-1}) = \delta(0, a^{-1}).$$

Consequently $y^k x^k = (e_0 + e_{-k}, 1)$ for every integer k. All relators of P_n hold in G_n, including $(r_0 r_2)^2 = 1$, so the marking gives a homomorphism $\Phi : P_n \to G_n$.

The common cyclic-recognition corollary applies to these exact relations. It constructs the inverse by

$$(e_0 + e_i, 1) \longmapsto y^{-i} x^{-i}, \qquad (0, a) \longmapsto x, \qquad (0, b) \longmapsto b.$$

It is well defined on indices modulo n, and the common proof verifies the conjugation relations needed for a semidirect-product homomorphism. The composites fix all translation basis vectors and dihedral generators, and all generators of P_n, respectively. Thus Φ is an isomorphism. Together with the preceding section, this proves Theorem 1 without a finite-order upper bound.

4 Exact identifications and parameters

The source uses the ordered basis

$$f_j = e_0 + e_{n-j} \qquad (1 \leq j < n)$$

of E_n. Direct application of the two permutations gives

$$bf_j = f_{n-j}, \qquad af_1 = f_{n-1}, \qquad af_j = f_{j-1} + f_{n-1} \quad (2 \leq j < n).$$

These are exactly its matrices V, A; the matrix of $s = ab$ is $U = AV$, and its last coordinate vector is $f_{n-1} = q$. This verifies the literal matrix marking in [LZ25, Section 3], in the consistent left-action convention used here.

For the earlier geometric construction, use the standard marking $(e_0, 1), (0, s), (0, b)$ of the generalized cube over the n-gon in $\mathbb{F}_2^\Omega \rtimes H$. Endpoint halving changes the first marker to

$$(e_0, 1)(0, s)(e_0, 1) = (q, s)$$

and leaves $(0, s), (0, b)$. Ordering the new three-node diagram as a string and reversing it gives (1). Thus this is the same distinguished group, not merely another group of the same order. The halving theorem and polygon count are proved in the common paper and the companion application to generalized cubes; the direct intersection proof above also establishes this application's geometric conclusion independently of that construction theorem.

Every odd $n \geq 3$ gives [LZ25, Corollary 3.5]. More generally, putting $n = 2^s p \geq 3$, for integers $p \geq 2$, $s \geq 0$, gives

$$|G_n| = 2^{2^s p + s} p.$$

This includes the prime range of its Corollary 3.6 and also all composite $p \geq 2$. The value $(p, s) = (2, 0)$ requires a separate construction: C_2^3 with its three coordinate involutions is a string C-group of type $\{2, 2\}$, has eight flags and admits reversal of the coordinates. No faithful permutation action of D_4 on two vertices is assumed.

5 Conventions, attribution and formal scope

The multiplication displayed in [LZ25, physical page 7] uses $\varphi(g)^{-1}w$ while describing φ as a homomorphism acting on column vectors. With left composition this is not generally associative. For $n = 3$, take $u = (0, a)$, $v = (0, b)$ and $w = (e_0 + e_1, 1)$. That rule gives translation coordinate $e_0 + e_1$ in $(uv)w$ and $e_1 + e_2$ in $u(vw)$. Our multiplication uses $\varphi(g)w$. A right-action convention is also possible, but its composition order must change consistently. The recovered principal statements do not assert the literal validity of that source display.

The original existence and presentation results remain attributed to the two cited papers. The common displacement-commutativity proof expands the argument of Li–Zhang. The present proof change is the use of the common augmentation-module recognition theorem to construct the inverse explicitly; the previous normal-subgroup order bound is no longer required. No priority claim is made for that general method.

The existing file `BinaryDihedralDisplacements.lean` verifies the integer displacement identity, its commutative closure and the finite-period adapter, with five reported declarations using only standard axioms. The companion file `SemidirectRecognition.lean` proves the generic inverse construction from actual component homomorphisms, covariance and generation, with nine reported declarations. The augmentation basis, its cyclic instantiation here, intersection property, matrix identification and geometric correspondence remain written proofs. Neither collection is a complete formalization of Theorem 1. The former finite model program is retained as an optional diagnostic, not a premise for the quantified theorem.

References

[U] *Changes of chamber generators and binary translation groups*, the common paper of Topic C.

[MW20] A. Montero and A. Ivić Weiss, *Locally spherical hypertopes from generalised cubes*, arXiv:2001.11082v1, 29 January 2020.

[LZ25] M. Li and W.-J. Zhang, *Regular 3-polytopes of type* $\{n, n\}$, arXiv:2505.09505v1, 14 May 2025. Theorems 1.3 and 3.4, Corollaries 3.5–3.6.

[MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section 2E.